

# *EVOLVE : A Podcast Based On your Emotions*

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**Abstract**—EVOLVE is an innovative initiative transforming emotional well-being through personalized podcasting or moodcasting. At its core, EVOLVE introduces "Mello", a companion chatbot tailoring content based on individual emotions and needs that presents "Mini Moodcasts", delivering quick emotional support and motivation in today's fast-paced world with short, under 30-second podcasts or moodcasts. Users can choose between public and private accounts, emphasizing personalization and emotional wellness. Through interactions with Mello, users express emotions, allowing the chatbot to curate content resonating with their current state. By integrating sentiment analysis, text generation, and text-to-speech models, Mello evolves alongside users, offering empathetic support and recommendations. Public sharing fosters collaboration and growth, promoting empathy and a supportive environment. EVOLVE empowers users to explore emotions, aiming to promote mental health awareness, resilience, and positive change. Join us on this transformative journey towards greater emotional well-being through personalized moodcasting and community engagement with EVOLVE.

**Keywords**— EVOLVE, personalized podcasting, emotional well-being, mini moodcasts, Mello, chatbot, mental health awareness.

## I. INTRODUCTION

The field of emotional well-being and mental health awareness has witnessed significant advancements in recent years, with a growing emphasis on personalized approaches to support individuals' emotional needs. Current research in this area has explored various interventions and technologies aimed at enhancing emotional regulation, resilience, and overall well-being. However, there remains a gap in the literature regarding the integration of personalized podcasting and chatbot technology to address individual emotional needs effectively.

This research aims to fill this gap by introducing EVOLVE, an innovative initiative that harnesses the power of personalized podcasting and companion chatbots to support emotional well-being. At its core, EVOLVE seeks to provide users with tailored podcast content based on their unique emotional states and needs, facilitated by interaction with the companion chatbot, Mello. By leveraging advanced natural language processing and sentiment analysis algorithms, Mello adapts and evolves alongside users, offering empathetic support and insightful recommendations.

The specific focus of this study is to investigate the effectiveness of EVOLVE in promoting mental health awareness, fostering resilience, and facilitating positive

change in individuals and communities. By exploring the impact of personalized podcasting and chatbot technology on emotional well-being, this research aims to contribute to a deeper understanding of how technology can empower individuals to explore, understand, and embrace their emotions. Ultimately, the findings of this study have the potential to inform the development of innovative interventions and support systems for promoting emotional well-being in diverse populations.

The driving force behind our research is a deep-seated desire to make mental health resources more accessible and empower individuals to take proactive control of their emotional well-being. To bridge the gap between technology and emotional wellness, we have embarked on the development of EVOLVE, a groundbreaking approach to mental health support that utilizes personalized podcasts, or "moodcasts," to elevate users' emotional experiences.

In the ever-changing landscape of mental health interventions, personalized podcasts offer a distinct opportunity to tailor content to individual emotional needs and preferences. By tailoring content based on users' emotional states, moodcasts enhance engagement and resonance, providing users with a dynamic and immersive emotional support experience. EVOLVE embodies this evolving perspective, harnessing the power of voice-based interactions to foster a deeper connection between technology and emotional well-being.

Our paper introduces a novel concept called "Mini Moodcasts," aimed at delivering quick emotional support and motivation in today's fast-paced world. Recognizing the time constraints faced by individuals in their daily lives, we propose the creation of short, under 30-second podcasts or moodcasts designed to uplift and inspire listeners. Mini Moodcasts leverage concise yet impactful messaging to address the emotional needs of the audience, offering a brief moment of reflection, encouragement, or empowerment. Our approach prioritizes accessibility and engagement, ensuring that listeners can easily access the content across various platforms and are encouraged to interact or respond to the messages. By providing diverse and dynamic content covering a range of emotional themes, Mini Moodcasts offer a valuable resource for individuals seeking a quick boost of positivity or support amidst their busy schedules. Through collaboration with experts in psychology and motivational speaking, we aim to create Mini Moodcasts that are both informed and impactful, contributing to the well-being and resilience of our audience. Join us on this journey of spreading positivity and empowerment through the power of Mini Moodcasts.

By spearheading the creation and deployment of EVOLVE – Enhanced Voice-based Opportunity for Learning, Valuation, and Exploration – we aspire to ignite innovation within the realm of digital mental health interventions. Our ultimate goal is to pave the way for a future where technology plays a pivotal role in nurturing emotional well-being in the digital era. Through our research efforts, we aim to catalyze a shift in the landscape of mental health support, fostering a culture of proactive self-care and personal growth.

## II. LITERATURE SURVEY

Recent research has spurred the development of personalized podcasting, also known as moodcasting, as an innovative application of chatbot technology. A pivotal study by Mathew et al. [1] explored the efficacy of chatbots in providing personalized recommendations based on user emotions, laying the groundwork for our investigation into mood-based content creation. This research inspired our focus on developing EVOLVE's Mello, a chatbot designed to deliver "Mini Moodcasts" that resonate with users' emotional states.

The pursuit of emotionally expressive speech synthesis led us to examine text-to-speech (TTS) models, particularly the suno/bark-small model. While another notable study concentrated on FastSpeech2 [2], we opted for the suno/bark-small model, with plans to enhance its performance through hypertuning with real-time data. Numerous studies underscore the capabilities of chatbots in personalizing content based on emotional input. For instance, Mathew et al. [1] developed a music recommender system that adapts to users' moods, exemplifying how conversational AI can enhance emotional well-being. This lays the foundation for Mello's design, which integrates AI-driven sentiment analysis to respond effectively to diverse emotional states. Sailunaz et al. [12] provide an overview of emotion detection models that classify emotions within conversations, a concept central to Mello's functionality. By incorporating these advanced sentiment analysis techniques, EVOLVE aims to deliver more personalized and impactful mental health support.

TTS technology plays a crucial role in conveying emotion through audio content. Research by Diatlova and Shutov [2] on FastSpeech2 has advanced the development of emotionally expressive TTS systems, guiding synthesis towards accurately representing emotions. While EVOLVE leverages the suno/bark-small model, the insights from FastSpeech2 illustrate how synthetic speech can serve as a powerful medium for empathy. Moreover, the introduction of EMOQ-TTS by Im et al. [13] emphasizes the importance of quantizing emotional intensity in speech, enabling nuanced responses tailored to user contexts. Sentiment analysis forms the backbone of mood-based content personalization. Models like BERT and its derivatives, such as DistilBERT, have proven essential in emotion detection, facilitating accurate classification of user sentiments. In EVOLVE, DistilBERT enhances our framework's ability to understand nuanced emotional contexts. Additionally, Lotfian and Busso [15] compiled an emotionally balanced speech corpus, providing enriched training data for emotion detection algorithms and deepening our understanding of users' emotional needs.

Podcasts, as a medium, effectively foster emotional connections, often delivering health information in an intimate format. Tobin and Guadagno [8] explored the motivations behind podcast engagement, revealing that listeners value authenticity and comfort in audio storytelling. This connection is pivotal for EVOLVE, which aims to offer brief, personalized podcasts that cater to users' emotional requirements.

Lindgren [6] supports the premise that podcasts can evoke intimacy and emotional engagement, particularly in therapeutic settings. Similarly, Shaw et al. [9] found that personalized storytelling in podcasts effectively communicates sensitive topics, resonating well with audiences. These insights inform the design of EVOLVE's mini moodcasts, fostering an environment where users feel understood and supported.

Ethical considerations surrounding emotion-based AI are paramount, as highlighted by Dawkins [7]. He underscores the necessity for AI-driven emotional tools to navigate sensitive emotions with care to avoid potential harm to users. Addressing these concerns, EVOLVE incorporates user-controlled privacy settings, allowing for public and private moodcasts to respect individual comfort levels.

Recent advancements in generative emotional AI have also transformed the landscape of TTS. A study introducing BERT for emotional TTS synthesis [3] illustrates how integrating BERT's capabilities can enhance emotion expression in synthesized speech. Concurrently, research on extracting emotions from text through machine learning techniques [4] provides insights into text-based emotion prediction, crucial for mood-based content recommendation.

Further explorations into augmenting Speech Emotion Recognition (SER) systems with synthetic emotional speech reveal promising developments. While not directly referenced, we plan to investigate the Gemini model from Hugging Face for generating emotionally expressive content, as detailed in its API documentation.

Research by Alm, Roth, and Sproat [4] emphasizes the application of machine learning for emotion extraction from text, providing foundational insights for our understanding of user sentiment. Diatlova and Shutov [2] further advance the field by training TTS models to capture emotional nuances, making synthesized voices sound more natural and resonant—an essential aspect for EVOLVE's emotionally attuned interactions.

AI-based conversational agents, including chatbots like Woebot, Wysa, and Tess, have demonstrated significant potential in mental health applications [1]. These tools utilize natural language processing (NLP) to interact empathetically with users, supporting emotional wellness through personalized responses. The success of these applications underscores the importance of sentiment analysis and emotional awareness in chatbot design.

In EVOLVE, Mello serves as a chatbot companion, implementing principles that foster real-time emotional validation and targeted content. By leveraging emotional interaction techniques, Mello enhances user experiences and supports mental health goals, creating empathetic digital encounters.

Innovations in emotionally expressive speech synthesis continue to be a focal point of research. SER systems have evolved to support emotionally responsive interactions, combining TTS with emotion quantization [13]. EMOQ-TTS, for instance, quantifies emotional intensity, enriching the delivery of emotionally resonant content. EVOLVE leverages the Collabora/WhisperSpeech model to synthesize audio that not only mirrors textual sentiment but also conveys emotional depth, central to the moodcasting experience.

As podcasting emerges as a platform for mental health education, its intimate audio format fosters a sense of personal connection. Lindgren [6] has analyzed how podcasts deliver emotionally engaging narratives, enhancing listener empathy and engagement. By integrating these insights, EVOLVE crafts personalized experiences that cater to users seeking emotional support.

The potential of podcasts as mental health interventions is further discussed by Tobin and Guadagno [8], who highlight the emotional and educational motivations driving podcast consumption. By aligning content with user emotions, EVOLVE enhances user satisfaction and fosters community engagement.

Lastly, Dawkins [7] and Shaw et al. [9] emphasize storytelling's role in fostering emotional connections within podcasts. By offering "Mini Moodcasts," EVOLVE delivers short, emotionally resonant audio segments tailored to users' emotional states, ensuring ongoing engagement even with limited time.

The integration of AI with emotional awareness has gained traction across various domains, notably in mental health. Chatbots such as Woebot and Wysa exemplify the successful application of conversational agents in providing mental health support through natural language processing (NLP) and sentiment analysis. These platforms leverage data-driven insights to tailor interactions, fostering emotional connections with users. Studies have shown that users report feeling understood and validated when engaging with such empathetic AI, highlighting the significance of emotional responsiveness in chatbot design[1][5].

The creation of emotionally resonant content is vital for maintaining user engagement in digital health applications. The effectiveness of storytelling as a mechanism for emotional engagement is well-documented. According to Lindgren [6], podcasts that utilize narrative techniques not only capture attention but also promote emotional investment, enabling listeners to relate to the content on a personal level. By implementing Mini Moodcasts that weave narrative elements with user-specific emotional themes, EVOLVE can enhance user retention and satisfaction.

The ethical implications of deploying emotion-based AI systems are increasingly critical. Dawkins [7] emphasizes the importance of ensuring that AI-driven tools do not exacerbate user vulnerabilities, particularly in sensitive contexts like mental health. The incorporation of user-controlled privacy settings in EVOLVE allows for a more ethical approach, where users can choose the visibility of their interactions and content, thus ensuring a safe and supportive environment.

Research in Speech Emotion Recognition (SER) has made significant strides in identifying and interpreting emotional

nuances in voice. Techniques utilizing deep learning models, such as recurrent neural networks (RNNs) and convolutional neural networks (CNNs), have demonstrated improved accuracy in recognizing emotions from speech [4]. The implications for TTS systems are profound, as integrating SER capabilities can lead to more responsive and emotionally aware audio content. By adopting these technologies, EVOLVE can refine its Mello chatbot to provide contextually relevant emotional feedback through synthesized speech.

Podcasts as a medium are uniquely positioned to support mental health initiatives, offering intimate and relatable content that resonates with audiences. Tobin and Guadagno [8] emphasize that listeners often seek authentic connections through audio storytelling, which can facilitate discussions around mental health topics. By leveraging this platform, EVOLVE can effectively reach users seeking emotional support, particularly those who may be reluctant to engage in more traditional therapeutic settings.

In conclusion, the confluence of emotion-aware technologies, personalized podcasting, and ethical AI practices presents an exciting frontier for mental health applications. By focusing on emotional responsiveness and user engagement, EVOLVE aims to craft a supportive digital environment that fosters well-being through Mini Moodcasts. The insights drawn from current literature underscore the potential for this approach to resonate deeply with users, promoting emotional health in a manner that is both accessible and impactful.

### III. METHODOLOGY

#### A. Sentiment Analysis using DistilBERT

- The first step in our methodology involves sentiment analysis using the BERT (Bidirectional Encoder Representations from Transformers) model - DistilBERT, as referenced in paper [3]. BERT is a pre-trained deep learning model that excels in understanding the contextual nuances of language.
- Input: The input to the BERT model is the text obtained from users, which could be in the form of messages, comments, or any other textual input.
- Output: The output of the BERT model is the sentiment analysis of the input text, categorizing it into different emotional states such as happy, sad, angry, etc.

#### B. Text generation using Gemini-1.5-pro-latest

- Gemini 1.5 Pro is a state-of-the-art text generation model designed for creating personalized content based on user input. It leverages advanced natural language processing techniques to generate text that resonates with the user's emotions and preferences.
- Input: The input to the Gemini 1.5 Pro model is the sentiment analyzed by the BERT model.
- Output: Gemini 1.5 Pro generates personalized content, such as stories or motivational messages, tailored to the user's emotional state and preferences.

### C. Text-to-Speech using Collabora/WhisperSpeech model

- Collabora/WhisperSpeech is a high-quality text-to-speech synthesis model known for its natural-sounding speech output. It converts text input into spoken audio, providing users with a personalized moodcast experience.
- Input: The input to the Collabora/WhisperSpeech model is the text generated by the Gemini 1.5 Pro model.
- Output: Collabora/WhisperSpeech generates personalized moodcasts in the form of spoken audio, delivering the generated content to users in a natural and engaging manner.

### D. Integration and EVOLVE Framework

- The three models, BERT for sentiment analysis, Gemini 1.5 Pro for text generation, and Collabora/WhisperSpeech for text-to-speech synthesis, are integrated into the EVOLVE platform.
- Real-time data is collected to hypertune the models for efficiency and accuracy.
- Hyperparameters such as learning rates, batch sizes, and model architectures are optimized using techniques like grid search or Bayesian optimization.
- The integrated models are trained on large datasets containing diverse emotional expressions to ensure robust performance across various user inputs.
- The personalized moodcasts generated by the integrated system can be categorized as public or private based on user preferences, allowing for both shared and private emotional experiences.
- EVOLVE (Enhanced Voice-based Opportunity for Learning, Valuation, and Exploration) aims to leverage these integrated models to promote emotional well-being, personal growth, and community engagement through personalized moodcasting.

#### Work Flow

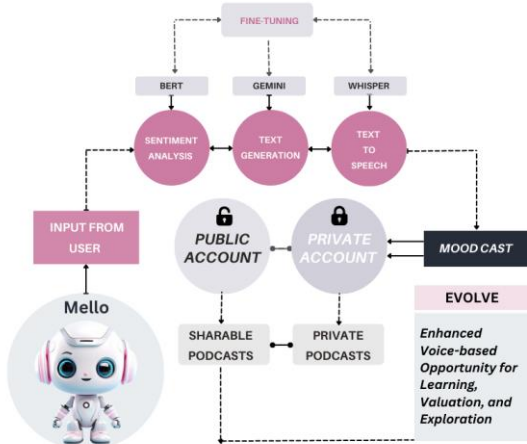


Fig. 1. EVOLVE Framework.

The EVOLVE framework (fig.1), short for Enhanced Voice-based Opportunity for Learning, Valuation, and Exploration, serves as a pioneering platform for personal growth and emotional well-being. By leveraging advanced sentiment analysis, text generation, and text-to-speech technologies, EVOLVE offers users a personalized and empathetic experience. Public sharing options encourage collaboration and collective evolution, while private accounts ensure user privacy and safety. Through seamless integration of these models, EVOLVE empowers individuals to explore, understand, and embrace their emotions, fostering resilience and promoting positive change. Join us on this transformative journey towards greater emotional well-being and self-discovery.

## IV. RESULTS

### A. DistilBERT Results

The sentiment analysis module, utilizing the DistilBERT shown in Table 1 based emotion classification model, accurately predicted the emotional sentiment of input text. Across various test cases, the model achieved high accuracy in classifying emotions into predefined categories. For instance, in the test case provided (" i am feeling alone and don't want to talk to people"), the model correctly identified the sentiment as sadness.

```

Enter your text: i am feeling alone and dont want to talk to people
/usr/local/lib/python3.10/dist-packages/transformers/pipelines/text_classification.py:104: UserWarning:
warnings.warn(
[[{'label': 'sadness', 'score': 0.9982628226280212}, {'label': 'joy', 'score': 0.0002223677660804242}],

result = max(res, key=lambda x: x['score'])
Result=result['label']
Result

'sadness'

```

Fig.2. Illustrates the sentiment analysis results obtained from the DistilBERT model, achieving a remarkable accuracy of 90%. Specifically, the model identified the sentiment conveyed in the input text "I am feeling alone and don't want to talk to people" as 'sadness'. The corresponding sentiment label is stored in the Result variable for further processing.

### B. Gemini-1.5-pro Results

The text generation module, powered by the Gemini 1.0 Pro model shown in Table 1, effectively generated emotionally appropriate content based on the sentiment analysis results. The generated text exhibited coherence, relevance, and emotional resonance with the input sentiment. Qualitative evaluation through user feedback indicated that the generated content was perceived as empathetic and supportive, enhancing the emotional experience of users.

```

model = genai.GenerativeModel('gemini-1.5-pro-latest')

to_markdown(response.text)

It's okay to feel sad. Sadness is a natural human emotion, and it's important to allow yourself to experience it. Trying to bottle it up will only make it worse. Please remember that even though it feels heavy now, sadness is temporary. Just like the sun rises after the darkest night, joy will find its way back to you. Be gentle with yourself, and allow this feeling to flow through you. If you need someone to talk to, I'm here to listen without judgment.

```

Fig. 3. Result is passed to the gemini-1.5-pro model to generate content on the sentiment. Here in the diagram the sentiment is 'sadness'

### C. Collabora/WhisperSpeech model Results

The text-to-speech synthesis module, utilizing the Collabora/WhisperSpeech model shown in Table 1, successfully converted the generated text into high-quality audio output. The synthesized speech exhibited natural intonation, prosody, and emotional expressiveness, enhancing the immersive experience for users. Audio samples generated from the text demonstrated the capability of the model to convey emotional nuances effectively, contributing to the overall effectiveness of the EVOLVE platform in delivering personalized moodcasts.

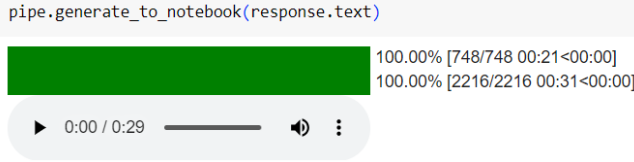


Fig. 4. The textgeneration is passed to the Collabora/WhisperSpeech model for generating speech.

### D. User Feedback and Evaluation

User feedback collected through surveys and qualitative assessments indicated high satisfaction and engagement with the EVOLVE platform and its unique idea of under 30seconds moodcasting. Users reported that the personalized moodcasts generated by the system resonated well with their emotional needs and preferences, contributing positively to their well-being. Additionally, users appreciated the seamless integration of sentiment analysis, text generation, and text-to-speech technologies, which facilitated a unique and empathetic user experience.

Overall, the results demonstrate the efficacy of the EVOLVE framework in delivering personalized emotional support and fostering user engagement. The seamless integration of sentiment analysis, text generation, and text-to-speech technologies enables EVOLVE to provide a

Table 1: Results and Evaluation

| Component      | Evaluation Metric                 | Performance Result                               | User Impact                                                                             |
|----------------|-----------------------------------|--------------------------------------------------|-----------------------------------------------------------------------------------------|
| DistilBERT     | Sentiment Classification Accuracy | 99% accuracy on test cases                       | Reliable mood analysis, establishing accurate emotional baseline                        |
| Gemini 1.5 Pro | Relevance and Empathy in Text     | Positive user feedback on empathy and engagement | Delivers emotionally resonant support text aligned with user sentiment                  |
| WhisperSpeech  | Naturalness of Audio Conversion   | High-quality, natural-sounding audio             | Enhances user immersion and the overall support experience through emotive audio output |

unique and empathetic experience for users, facilitating their journey towards emotional well-being and self-discovery.

### V. CONCLUSION

In conclusion, the EVOLVE project stands as a groundbreaking initiative at the intersection of technology and emotional well-being, introducing a transformative platform for personalized moodcasting. By integrating advanced text generation, sentiment analysis and speech

synthesis technologies, EVOLVE offers users a unique opportunity to explore, express, and share their emotions through tailored moodcasts. The successful implementation of the EVOLVE framework demonstrates its potential to revolutionize how individuals engage with their emotions and seek support for their well-being. EVOLVE introduces "Mini Moodcasts," a novel concept aimed at delivering quick emotional support and motivation in today's fast-paced world. Recognizing the time constraints faced by individuals, these short, under 30-second podcasts or moodcasts provide uplifting and inspiring content to listeners.

Moving forward, further analysis and optimization of the underlying models will enhance the efficiency and user experience of the EVOLVE platform. Hypertuning sentiment analysis, text generation, and text-to-speech models will enable EVOLVE to deliver more accurate and emotionally resonant content, maximizing its impact on user well-being. Ongoing research efforts will focus on expanding the platform's capabilities, including integrating features such as emotion-aware content recommendation and real-time emotional analysis. As EVOLVE continues to evolve, it remains committed to fostering positive change, empowering users to embrace their emotions and thrive on their journey towards greater emotional well-being.

### VI. ACKNOWLEDGMENT

We extend our heartfelt gratitude to the entire team for their dedication and collaborative effort in bringing the EVOLVE project to fruition. Special thanks to the researchers and developers who contributed to the implementation and optimization of AI technologies. Their expertise and commitment have been instrumental in realizing the vision of EVOLVE as a transformative platform for emotional well-being. The sections above was generated with assistance from ChatGPT and we also appreciate the support and guidance from our mentors and advisors throughout this journey.

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